AI Assisted Coding

Assignment – 16.3

Name: A.sai shrehan

Rollno:2303A52299

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Task1:

Create schema and queries for a Hospital Management System.

Instructions:

• Tables: Doctors, Patients, Appointments.

• Use AI to define constraints (unique IDs, valid dates).

• Generate queries:

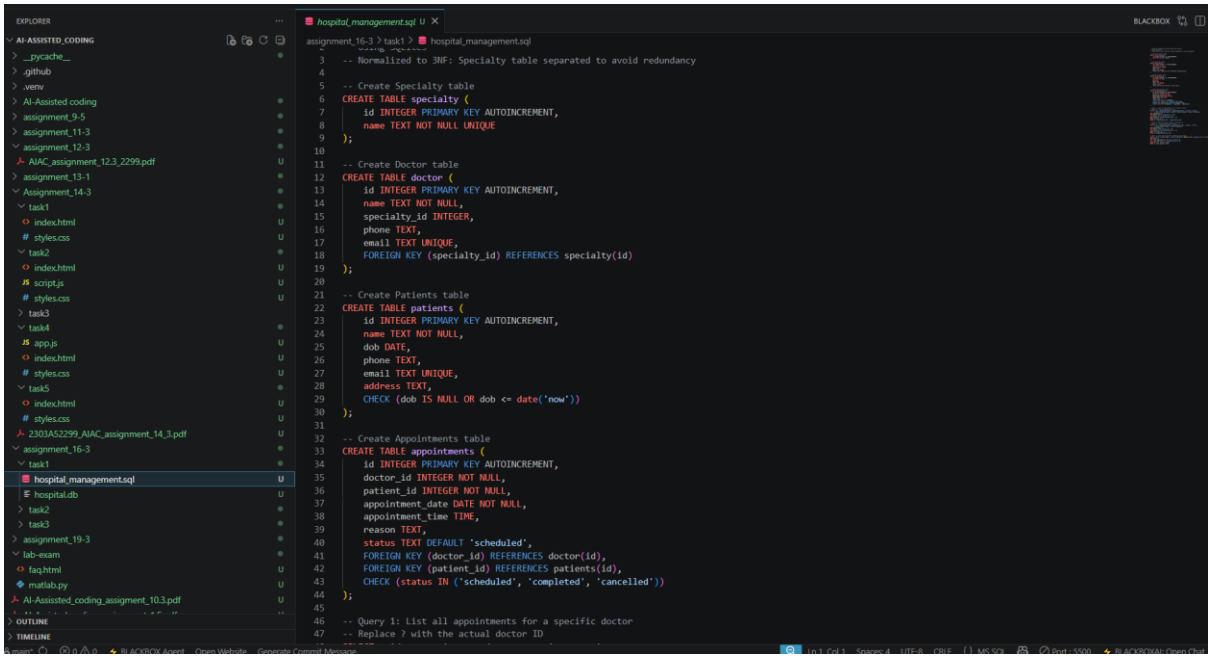
• List all appointments for a specific doctor.

• Retrieve patient history by patient ID.

• Count total patients treated by each doctor.

Expected Output:

• Normalized schema and SQL queries with joins.



Task 2 – Real-Time Application: Online Shopping Database

Scenario:

Design a database for an E-commerce platform.

Requirements:

• Tables: Users, Products, Orders, OrderDetails.

• Generate queries to:

• Retrieve all orders by a user.

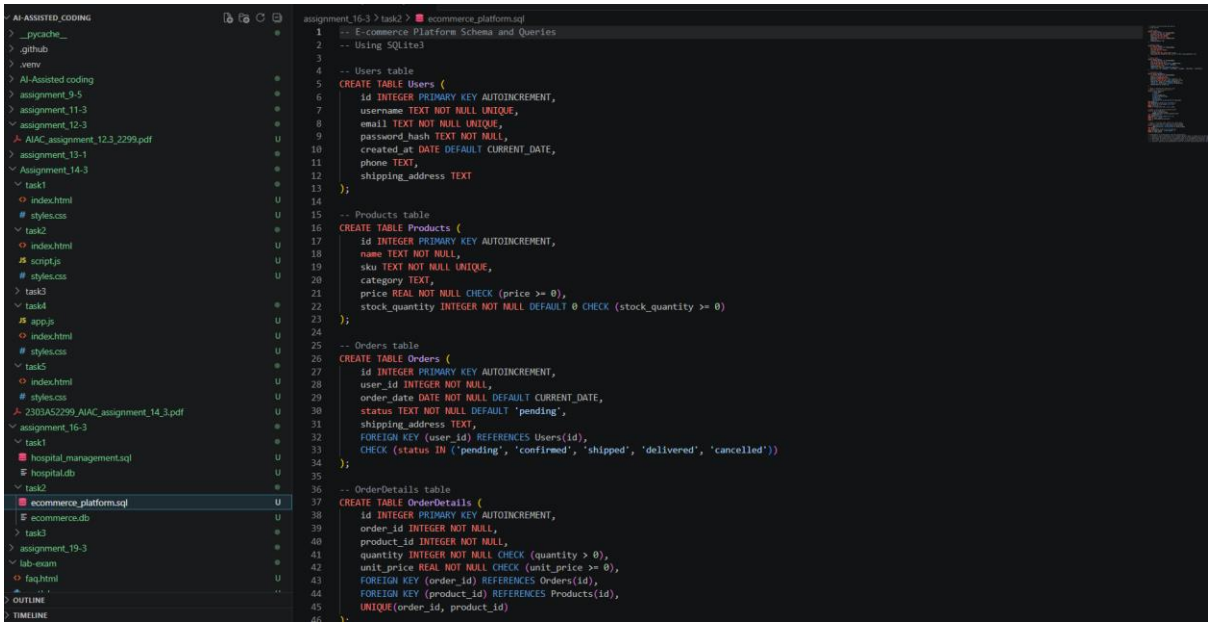
• Find the most purchased product.

• Calculate total revenue in a given month.

• AI should also suggest normalization improvements and query optimization.

Expected Output:

• Complete schema with relationships + SQL queries for analytics.



Task 3 – Library Database

Task:

Design schema for a Library Management System.

Instructions:

• Tables: Books, Members, Loans.

• Use AI to suggest indexing strategy for faster queries.

• Generate queries:

• Retrieve all books currently issued.

• Find overdue books (loan date > 30 days).

• Count number of books loaned by each member.

Expected Output:

• Schema + SQL queries demonstrating joins and conditions.

library_system.sql U

assignment_16-3 > task3 > library_system.sql

```sql
-- Library Loan System Schema and Queries
-- Using SQLite3

-- Books table
CREATE TABLE Books (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    title TEXT NOT NULL,
    author TEXT,
    isbn TEXT UNIQUE,
    published_date DATE,
    available_copies INTEGER NOT NULL DEFAULT 1 CHECK (available_copies >= 0)
);

-- Members table
CREATE TABLE Members (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT NOT NULL,
    email TEXT UNIQUE,
    phone TEXT,
    membership_date DATE NOT NULL DEFAULT CURRENT_DATE
);

-- Loans table
CREATE TABLE Loans (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    book_id INTEGER NOT NULL,
    member_id INTEGER NOT NULL,
    loan_date DATE NOT NULL DEFAULT CURRENT_DATE,
    return_date DATE,
    status TEXT NOT NULL DEFAULT 'issued',
    FOREIGN KEY (book_id) REFERENCES Books(id),
    FOREIGN KEY (member_id) REFERENCES Members(id),
    CHECK (status IN ('issued', 'returned', 'lost'))
);

-- Indexing strategy for faster queries:
-- 1. Create an index on Loans(book_id) for book-based lookups.
-- 2. Create an index on Loans(member_id) for member history and aggregations.
-- 3. Create an index on Loans(status, loan_date) to speed up currently issued and overdue queries.
-- 4. Create an index on Books(isbn) for fast book lookup by ISBN.

CREATE INDEX idx_loans_book_id ON Loans(book_id);
CREATE INDEX idx_loans_member_id ON Loans(member_id);
CREATE INDEX idx_loans_status_loan_date ON Loans(status, loan_date);
CREATE INDEX idx_books_isbn ON Books(isbn);
```